

Exercise 2: E-commerce Platform Search Function

Aim

To implement Linear Search and Binary Search algorithms in Java and compare their performance for searching products in an e-commerce platform.

Problem Understanding

Searching is one of the most common operations in an e-commerce application. The efficiency of the search algorithm directly affects the user experience. Big O notation is used to measure the performance of algorithms as the input size increases.

Big O Notation

Big O notation describes how the execution time of an algorithm grows with the size of the input.

Search Scenarios

- **Best Case:** The element is found immediately.
- **Average Case:** The element is found somewhere in the middle.
- **Worst Case:** The element is found at the end or is not present.

Classes Used

Product

Stores product information:

- Product ID
- Product Name
- Category

SearchFunction

Implements:

- Linear Search
- Binary Search

Time Complexity

Algorithm Best Case Average Case Worst Case

Linear Search $O(1)$ $O(n)$ $O(n)$

Binary Search $O(1)$ $O(\log n)$ $O(\log n)$

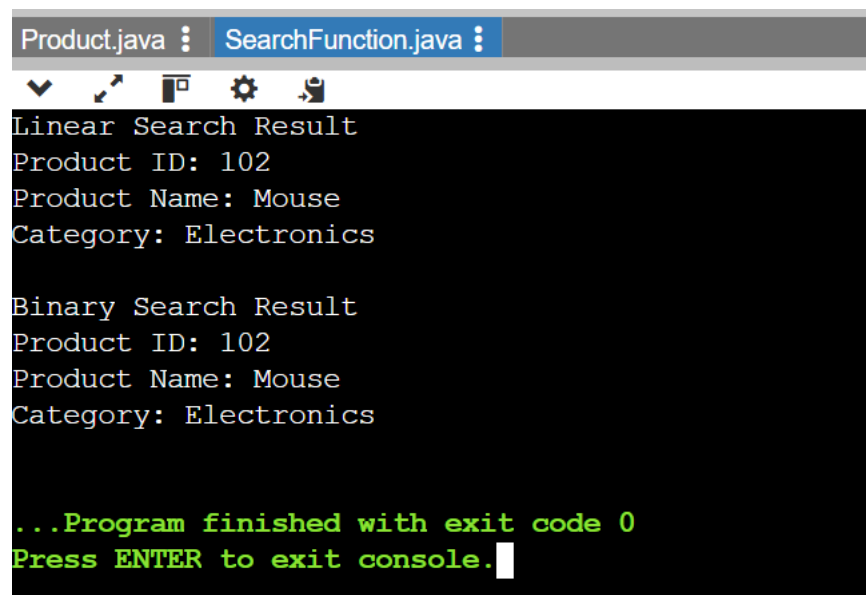
Analysis

Linear Search checks every element one by one and works on both sorted and unsorted data.

Binary Search requires the array to be sorted but performs much faster for large datasets by repeatedly dividing the search space into halves.

For an e-commerce platform with thousands of products, Binary Search is more efficient because it significantly reduces the search time.

Output



```
Product.java : SearchFunction.java :  
Linear Search Result  
Product ID: 102  
Product Name: Mouse  
Category: Electronics  
  
Binary Search Result  
Product ID: 102  
Product Name: Mouse  
Category: Electronics  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

Conclusion

This exercise demonstrates the implementation of Linear Search and Binary Search in Java and shows that Binary Search is more suitable for large, sorted datasets due to its $O(\log n)$ time complexity.